EEET2601 Engineering Computing 1

Arrays

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Problem

- To compute the average quiz marks of 3 students, we could do this:

```
int mark0, mark1, mark2;
printf("Enter mark0: ");
scanf("%d", &mark0);
printf("Enter mark1: ");
scanf("%d", &mark1);
printf("Enter mark2: ");
scanf("%d", &mark2);
float avg = (mark0 + mark1 + mark2) / 3.0;
```

IF / 3 -> INT devision

- What if we have 1,000 marks?

What is an array?



- Array is a collection of variables, array elements, of the same data type
- An array has *name, size*, and *index*
- Array size can't be changed after the array is created
- Array elements are stored in continuous memory locations access fast

How to create an array?

```
// Syntax
data_type array_name[array_size];


// Declare array, values of array elements are unknown
int marks[3];   Can't erase → need init


// Declare & initialize array
int marks[3] = {7, 4, 10};
int marks[]  = {7, 4, 10};
int marks[3] = {0};
```

Subscript operator

- Subscript operator "[index]" is used to refer to a specific array element
- We can calculate the average of five marks like this

```
float avg = (marks[0] + marks[1] + marks[2]) / 3.0;
```

- Or using a loop

```
float sum = 0; int size = 3;
for (int i = 0; i < size; i++) {
    sum += marks[i];
}
float avg = sum / size;
```

FLOAT / INT -> FLOAT

Array size known at runtime

```
// Create an array whose size only known at runtime
printf("Enter array size: ");
int size;
scanf("%d", &size);
int marks[size];

// Input marks
for (int i = 0; i < size; i++) {
    printf("Enter marks[%d]: ", i);
    scanf("%d", &marks[i]);
}
```

Array size known at run time

```
// Compute average mark
float sum = 0;
for (int i = 0; i < size; i++) {
    sum += marks[i];
}
float avg = sum / size;

// Output average mark
printf("The average mark is %f\n", avg);
```

Search for a value in an array

```
// Array of marks to search
int marks[] = {6, 2, 8, 9, 5};

// Input a value to search for
printf("Enter a value to search for: ");
int value;
scanf("%d", &value);
```

Search for a value in an array

```
// Search the array for the value
int found = 0;
...
```

```
// Output search result
if (found == 0) {
    printf("%d is not in the
                array\n", value);
} else {
    printf("%d is in the
                array\n", value);
}
```

Find the largest element of an array

```
// Array of marks
int marks[] = {6, 2, 8, 9, 5};

// Find the largest element of the array
int largest = marks[0];
...

// Output the largest element
printf("The largest element is %d\n", largest);
```

Sort an array in descending order

- Sort the following array: `int marks[] = {6, 2, 8, 9, 5};`
- Selection sort algorithm:
  1. Find the largest element from index 0 and swap it with the first element
  2. Find the largest element from index 1 and swap it with the second element
  3. Find the largest element from index 2 and swap it with the third element
  4. Find the largest element from index 3 and swap it with the fourth element
  5. The fifth element having index 4 must be already in the right order

Sort an array in descending order

```
// Sort array using selection sort algorithm
int marks[] = {6, 2, 8, 9, 5};
int tmp;
for (int i = 0; i < 5; i++) {
    for (int j = i + 1; j < 5; j++) {
        if (marks[j] > marks[i]) {
            tmp = marks[i];
            marks[i] = marks[j];
            marks[j] = tmp;
        }
    }
}
```

↑ can be 4 → last element should be sorted

Multi-dimensional array

```
// Syntax
data_type array_name[size1][size2] … [sizen];


// Examples
int chessboard[8][8] = {0};
int matrix[3][4] = {
    {72, 16, 22, 39},
    {44, 99, 65, 72},
    {84, 6, 100, 51}
};
```


swap
with only lvalues

Multi-dimensional array

```
// Find the largest element in the matrix
int largest = matrix[0][0];
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 4; j++) {
        if (matrix[i][j] > largest) {
            largest = matrix[i][j];
        }
    }
}
printf("Largest element is %d\n", largest);
```

Generate a random integer

- Use rand() and srand() in <stdlib.h>

```
// Set the random seed
srand(time(NULL));  // What happens if this line is removed?

// Generate a random integer
int num = rand();

printf("num is %d\n", num);
```

Generate an array of random integers

```
// Create array
int arr[5];

// Set the random seed
srand(time(NULL));

// Generate array of random integers and print them out
for (int i = 0; i < 5; i++) {
    arr[i] = rand();
    printf("%d ", arr[i]);
}
printf("\n");
```

References

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5. GeeksforGeeks, Selection Sort Algorithm, 2020.
6. TutorialsPoint, C Standard Library, 2020.